Guide to the Chain Rule

Derivatives of Composite Functions

1. Main Idea

The chain rule is used when a function is made by placing one function inside another function.

For example, in

$$y = (3x + 1)^5,$$

the expression $3x + 1$ is inside the power function.

We can think of this as

$$y = u^5 \qquad \text{where} \qquad u = 3x + 1.$$

The chain rule says that we differentiate the outside function first, keep the inside function unchanged, and then multiply by the derivative of the inside function.

2. Chain Rule Formula

If

$$y = f(g(x)),$$

then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x).$$

Equivalently, if

$$y = f(u) \qquad \text{and} \qquad u = g(x),$$

then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

3. Practical Steps

To use the chain rule:

1. Identify the inside function.
2. Identify the outside function.

3. Differentiate the outside function, leaving the inside unchanged.
4. Multiply by the derivative of the inside function.

4. Worked Examples

Example 1

Find the derivative of
$$y = (3x^2 - 5)^7.$$

The outside function is the power u^7, and the inside function is
$$u = 3x^2 - 5.$$

Differentiate the outside:
$$\frac{d}{du}(u^7) = 7u^6.$$

Differentiate the inside:
$$\frac{du}{dx} = 6x.$$

Therefore,
$$\frac{dy}{dx} = 7(3x^2 - 5)^6(6x).$$

So,
$$\boxed{\frac{dy}{dx} = 42x(3x^2 - 5)^6.}$$

Example 2

Find the derivative of
$$y = \sqrt{5x + 1}.$$

Rewrite the square root as a power:
$$y = (5x + 1)^{1/2}.$$

The outside function is $u^{1/2}$, and the inside function is
$$u = 5x + 1.$$

Then

$$\frac{dy}{dx} = \frac{1}{2}(5x+1)^{-1/2}(5).$$

Thus,

$$\boxed{\frac{dy}{dx} = \frac{5}{2\sqrt{5x+1}}.}$$

Example 3

Find the derivative of

$$y = \sin(4x^3).$$

The outside function is $\sin u$, and the inside function is

$$u = 4x^3.$$

The derivative of the outside function is

$$\frac{d}{du}(\sin u) = \cos u.$$

The derivative of the inside function is

$$\frac{du}{dx} = 12x^2.$$

Therefore,

$$\frac{dy}{dx} = \cos(4x^3)(12x^2).$$

So,

$$\boxed{\frac{dy}{dx} = 12x^2\cos(4x^3).}$$

Example 4

Find the derivative of

$$y = \cos^5(x).$$

This means

$$y = (\cos x)^5.$$

The outside function is u^5, and the inside function is

$$u = \cos x.$$

Then

$$\frac{dy}{dx} = 5(\cos x)^4(-\sin x).$$

So,

$$\boxed{\frac{dy}{dx} = -5\cos^4(x)\sin(x).}$$

Example 5

Find the derivative of

$$y = e^{2x^2+3x}.$$

The outside function is e^u, and the inside function is

$$u = 2x^2 + 3x.$$

Since

$$\frac{d}{du}(e^u) = e^u,$$

and

$$\frac{du}{dx} = 4x + 3,$$

we get

$$\frac{dy}{dx} = e^{2x^2+3x}(4x+3).$$

Thus,

$$\boxed{\frac{dy}{dx} = (4x+3)e^{2x^2+3x}.}$$

Example 6

Find the derivative of

$$y = \ln(7x - 2).$$

The outside function is $\ln u$, and the inside function is

$$u = 7x - 2.$$

The derivative of the outside function is

$$\frac{d}{du}(\ln u) = \frac{1}{u}.$$

The derivative of the inside function is

$$\frac{du}{dx} = 7.$$

Therefore,

$$\frac{dy}{dx} = \frac{1}{7x-2}(7).$$

So,

$$\boxed{\frac{dy}{dx} = \frac{7}{7x-2}.}$$

Example 7

Find the derivative of

$$y = \frac{1}{(x^2+4)^3}.$$

Rewrite the function using a negative exponent:

$$y = (x^2+4)^{-3}.$$

The outside function is u^{-3}, and the inside function is

$$u = x^2 + 4.$$

Then

$$\frac{dy}{dx} = -3(x^2+4)^{-4}(2x).$$

Thus,

$$\boxed{\frac{dy}{dx} = -6x(x^2+4)^{-4}.}$$

Equivalently,

$$\boxed{\frac{dy}{dx} = \frac{-6x}{(x^2+4)^4}.}$$

Example 8

Find the derivative of

$$y = \tan(5x^2 - 1).$$

The outside function is $\tan u$, and the inside function is

$$u = 5x^2 - 1.$$

Since

$$\frac{d}{du}(\tan u) = \sec^2 u,$$

and

$$\frac{du}{dx} = 10x,$$

we get

$$\frac{dy}{dx} = \sec^2(5x^2 - 1)(10x).$$

Therefore,

$$\boxed{\frac{dy}{dx} = 10x \sec^2(5x^2 - 1).}$$

Example 9

Find the derivative of

$$y = (\sin(3x))^4.$$

The outside function is u^4, and the inside function is

$$u = \sin(3x).$$

But the inside function also contains another inside function:

$$3x.$$

We apply the chain rule step by step:

$$\frac{dy}{dx} = 4(\sin(3x))^3 \cdot \frac{d}{dx}(\sin(3x)).$$

Now,

$$\frac{d}{dx}(\sin(3x)) = \cos(3x) \cdot 3.$$

Therefore,

$$\frac{dy}{dx} = 4(\sin(3x))^3 \cos(3x)(3).$$

So,

$$\boxed{\frac{dy}{dx} = 12\sin^3(3x)\cos(3x).}$$

Example 10

Find the derivative of

$$y = \ln\left((x^2+1)^5\right).$$

The outside function is $\ln u$, and the inside function is

$$u = (x^2+1)^5.$$

Using the chain rule:

$$\frac{dy}{dx} = \frac{1}{(x^2+1)^5} \cdot \frac{d}{dx}\left((x^2+1)^5\right).$$

Now differentiate the inside:

$$\frac{d}{dx}\left((x^2+1)^5\right) = 5(x^2+1)^4(2x).$$

Therefore,

$$\frac{dy}{dx} = \frac{5(x^2+1)^4(2x)}{(x^2+1)^5}.$$

Simplify:

$$\frac{dy}{dx} = \frac{10x}{x^2+1}.$$

Thus,

$$\boxed{\frac{dy}{dx} = \frac{10x}{x^2+1}.}$$

5. Summary

The chain rule is one of the most important rules for differentiation. It is used whenever a function is composed of another function.

The general rule is:

$$\boxed{\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)}$$

In words:

$$\boxed{\text{Differentiate the outside, keep the inside, then multiply by the derivative of the inside.}}$$

6. Practice Problems

Find the derivative of each function.

1. $y = (2x + 7)^6$
2. $y = \sqrt{x^2 + 9}$
3. $y = \sin(5x)$
4. $y = \cos(2x^2)$
5. $y = e^{x^3 - 4x}$
6. $y = \ln(x^2 + 6x)$
7. $y = (4x^2 - 1)^{-2}$
8. $y = \tan(3x + 1)$
9. $y = (\cos x)^3$
10. $y = \ln((x^2 + 4)^3)$